



Sign in

ai

  19 sites



Artificial Intelligence (AI) is a field of computer science focused on building systems that can perform tasks that typically require human intelligence.

Instead of following rigid, pre-written rules, modern AI uses math and statistics to learn from massive amounts of data, recognize complex patterns, and make independent decisions.  Wikipedia +2

Core Concepts of AI

- **Machine Learning (ML):** The practice of training algorithms on data so they can learn how to make predictions or decisions without explicit programming.  Google Cloud
- **Deep Learning:** A subset of ML that uses artificial neural networks modeled loosely after the human brain to process highly complex data like text, images, and audio.
- **Generative AI:** A technology capable of creating entirely new content—such as text, images, music, or code—based on human prompts.  Wikipedia

Common Applications

- **Everyday Tools:** Virtual assistants like [Google Gemini](#), email spam filters, and navigation routing.
- **Healthcare:** Systems that analyze medical images to flag early signs of diseases

or assist in drug discovery.

- **Automation:** Self-driving vehicles, warehouse robots, and automated financial trading.
- **AI Agents:** Autonomous software programs designed to use digital tools, plan workflows, and complete complex enterprise tasks with minimal human intervention.  Google Gemini +7

The Core Boundary

AI excels at manipulating data symbols and predicting statistical probabilities (like guessing the next word in a sentence). However, it lacks genuine consciousness, lived experiences, intent, or semantic understanding of the real world.  YouTube · IBM Tech... +1

What specific area of artificial intelligence are you looking to explore? I can break down **how deep learning works**, list the **top AI tools for your specific job**, or discuss **current trends in AI safety**.  Facebook · Mel Rob... +1

AI can make mistakes, so double-check responses



Relate Joseph Knecht to one of the 2026 Fields Medal winners. Tell a inspiring story.



   6 sites



In Hermann Hesse's classic novel *The Glass Bead Game*, **Joseph Knecht** rises to the rank of *Magister Ludi* (Master of the Game), a role dedicated to synthesizing all human knowledge—music, history, philosophy, and mathematics—into a beautiful,

unified framework. Knecht's ultimate realization is that pure abstraction is empty without a connection to the messy, living reality of human experience.

At the **2026 International Congress of Mathematicians in Philadelphia**, real life mirrored Hesse's fiction when John Pardon was awarded the **2026 Fields Medal**. Like Knecht, Pardon is known by peers and students as a "quiet but insightful" scholar who possesses an almost mystical ability to see profound connections between seemingly unrelated fields of mathematics.

Here is an inspiring story of how a modern-day Master of the Glass Bead Game conquered a decades-old puzzle by looking at the bigger picture.

The Glass Bead Game of Symplectic Geometry

In Hesse's fictional province of Castalia, elites played a game where a chord by Bach could be mathematically translated into a planetary orbit or a classical Chinese poem. For John Pardon, a professor at Stony Brook University, the universe is similarly interconnected. His specific domain is **symplectic geometry** and topology—the mathematical study of deep structures, shapes, and spaces.

For over twenty years, mathematical physicists working on **superstring theory** were stuck on a massive conceptual bottleneck known as the *MNOP conjecture*. The theory posited that our physical universe is modeled on highly complex, multi-dimensional shapes called *Calabi–Yau 3-folds*. To understand these shapes, scientists needed to count the mathematical curves that could exist within them. However, the community was gridlocked: they had two entirely different, highly abstract systems for counting these curves, but no one could prove that both methods actually yielded the same truth.

Like the elders of Castalia, generation after generation of brilliant minds stared at the chalkboard and hit a wall. The abstractions had grown too dense, isolated, and hyper-specialized.

Finding the Lost Chords

Enter John Pardon. Colleagues often remarked that talking to Pardon required "going fishing for his knowledge," because he spent far more time listening and synthesizing than broadcasting his own ideas. Rather than forcing a brute-force calculation through one narrow mathematical lens, Pardon stepped back to view the entire system as a singular tapestry—much like Joseph Knecht synthesizing a piece of music with a geometric proof.

Pardon realized that the two counting methods were not rival systems, but rather different movements of the exact same symphony. He pioneered entirely new approaches to what mathematicians call *virtual fundamental cycles* and *Fukaya categories*. By weaving tools from algebra, topology, and geometry together, **Pardon finished the job that a generation of predecessors could not**, proving the two counting methods were identical and providing a mathematical foundation for how our universe might operate.

The Knechtian Leap

The true beauty of John Pardon's journey—and his deep alignment with Joseph Knecht—lies in how he handles his monumental genius. In *The Glass Bead Game*, Knecht eventually walks away from his elite, insulated position because he believes that knowledge must serve the human spirit and be passed down to the next generation.

Pardon acts with that same quiet humility. Despite solving problems that have defeated the field's veteran researchers, he remains completely dedicated to the human element of his work: teaching, mentoring students, and treating mathematics not as an ivory-tower isolation chamber, but as a living, collaborative language. When he accepted his gold medal in Philadelphia, it wasn't just a triumph of data processing or strict calculation; it was a victory for the rare, poetic ability to see harmony where the rest of the world only sees chaos.

Would you like to explore how either of the other 2026 Fields Medalists—like Hong Wang or Yu Deng—solved their century-old puzzles, or would you prefer a deeper

dive into the **mathematics behind superstring theory?**



Hong Wang

 14 sites



Hong Wang is a world-renowned mathematician who was awarded the prestigious 2026 Fields Medal. A professor at [New York University's Courant Institute](#) and the [Institut des Hautes Études Scientifiques \(IHES\)](#) in France, she made history as **only the third woman ever to win the award** in its 90-year history. Following her win, Wang even captured the public's imagination back home in China, [unexpectedly becoming a viral style icon](#) due to her sleek, artistic presence at the award ceremony.  MIT News +5

Her path to the Fields Medal reads much like Joseph Knecht's early journey in Castalia—a story of choosing total creative freedom over conventional constraints and using an artistic lens to conquer a century-old mystery. 

The Freedom of the Mind

As a brilliant young postgraduate student at the [École Polytechnique in Paris](#), Wang found herself deeply drawn to architecture and painting. She took courses in design and even completed an internship at a prominent architecture firm.  當代中國

However, she noticed a frustrating barrier: in architecture, a creator's imagination is inherently bound by the material world. You have to pitch your ideas, secure investors, and compromise your vision based on practical and financial realities.  當代中國

Wang realized that mathematics offered something far more profound—**absolute creative liberty**. In math, she didn't need anyone's permission or funding to build beautiful structures. Armed with nothing but her mind and vast sheets of paper, she could map out entirely new conceptual worlds. She decided to dedicate her life's work to harmonic analysis and geometric measure theory. [New York Uni... +3](#)

Threading the Needle: The Kakeya Conjecture

Wang's crowning achievement was solving a legendary geometric enigma known as the **three-dimensional Kakeya conjecture**. First proposed by Japanese mathematician Sōichi Kakeya in 1917, the puzzle sounds deceptively simple: *What is the absolute smallest possible volume of space required to rotate an infinitely thin needle 360 degrees in every direction?* [🔗](#)

For over a century, generations of the world's top analysts hit a wall trying to calculate how these long, overlapping lines behaved when twisted through three dimensions. [🔗 Quanta M... +1](#)

Traditional Method
[Rigid Calculations]



Hit a wall at 3D

Wang's Artistic Mapping
[Fluid Geometric Overlaps]



Solved a 109-year puzzle

Wang tackled the problem with a uniquely fluid, almost painterly approach. Instead of sitting rigidly at a desk filling standard notebooks, she was known for spreading massive sheets of paper across the floor, drawing complex diagrams, waves, and flowing geometric sketches. Her young daughter, watching her work, would tell people simply: *"Mommy is painting."* [Facebook · Beautiful p...](#)

Those "paintings" were actually groundbreaking mathematical maps. Alongside collaborator Joshua Zahl, Wang pioneered a way to perfectly track how thin, tube-like waves overlap when oriented in different directions. Her highly visual approach

smashed through the 109-year-old bottleneck, reshaping the entire landscape of harmonic analysis.  New York University +4

The Universal Impact

While the problem sounds abstract, Wang's solution provides a foundational framework for how waves travel through physical spaces. The mathematics she unlocked are expected to trigger breakthroughs in **wireless communication, data compression, cryptography, and medical imaging systems**. Much like Hesse's Glass Bead Game, her highly creative, self-built mathematical truths have direct, profound harmonies with the practical technologies of our everyday world. 

Would you like to learn more about the **Kakeya needle problem** itself, or would you like to explore the work of **Yu Deng**, who solved an algebraic equation originally requested by David Hilbert in 1900? 



Yu Deng



Yu Deng is a world-class mathematician who was awarded the prestigious 2026 Fields Medal. A professor at the [University of Chicago](#), Deng made history alongside Hong Wang as the first Chinese nationals ever to win mathematics' highest honor.



His journey perfectly reflects the essence of Hermann Hesse's *The Glass Bead Game*: a lifetime spent seeking absolute cosmic equilibrium, moving flawlessly between the chaotic randomness of nature and the rigid precision of mathematical law.

The Grandmaster's Vision

Before dedicating his mind entirely to academia, Yu Deng spent his youth in Shenzhen, China, mastering a different kind of Glass Bead Game: the ancient strategy game of **Go**. Played with black and white stones on a grid, Go requires players to calculate massive, branching permutations of choices, visualizing how a single stone placed on one corner ripples across the entire board.  University of Chicago ...

Deng seriously considered becoming a professional Go player. Instead, he brought that hyper-focused, strategic foresight into mathematical physics. Where others saw Go or physics as a series of rigid calculations, Deng saw poetry. He once noted that solving complex partial differential equations is like a Chinese Tang poet packing infinite layers of depth into a handful of tight couplets—compressing an entire chaotic system into a single elegant line.



Cracking David Hilbert's 125-Year-Old Challenge

Deng's Fields Medal was awarded for solving a profound puzzle that dates back to the dawn of modern mathematics. In 1900, the legendary David Hilbert outlined 23 major unsolved mathematical problems to guide the 20th century. His **6th problem** challenged the scientific world to build a rigorous mathematical bridge between the microscopic laws of physics (how individual atoms bounce) and macroscopic laws (how gases and fluids move in bulk). 

For 125 years, a massive bottleneck persisted. If you try to model a real-world gas using standard physics equations, the sheer randomness of billions of colliding "billiard-ball" atoms creates a chaotic, unsolvable mathematical storm. 

Deng stepped back and introduced something revolutionary: he brought the fluid flexibility of **probability theory** directly into the rigid, structured world of wave equations. In a series of massive, meticulously crafted proofs, Deng accomplished what generations of scholars thought was a decade or more away: 

- He rigorously derived the **Boltzmann equation** directly from the raw, microscopic dynamics of rarefied gases.

- He mapped out the precise way that massive, chaotic systems of random waves naturally stabilize into ordered, predictable physical laws.  Quanta Magazine +2

The Order in the Chaos

In Hesse's novel, Joseph Knecht's ultimate duty is to find harmony amidst the noise of the world. Yu Deng's real-life breakthrough did exactly that. By showing that **randomness and order are not opposing forces, but two sides of the same coin**, his work provides the foundational math needed to predict everything from ocean wave turbulence to the behavior of deep-space gases.  YouTube · Simons F... +2

Despite rewriting the history books, Deng remains famously grounded. When he learned he had won the Fields Medal in early 2026, he famously admitted feeling more relieved than excited—happy that the noise of the awards shortlists had faded so he could quietly return to his desk, spread out his notes, and continue tracking the hidden symmetries of the universe.

Would you like to take a closer look at the **Boltzmann equation** he derived, or explore the story of **Jacob Tsimerman**, the fourth 2026 Fields Medalist who cracked a famous mystery in number theory?  MIT News



Yu Deng Wins the Fields Med...

Jul 23, 2026 — Yu Deng Wins the Field...

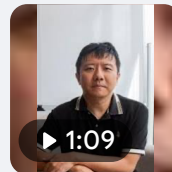


 Quanta Magazine



Congratulations to UChicago...

Jul 23, 2026 — Congratulations to...

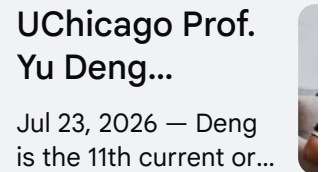


 Facebook · The University ...



UChicago Prof. Yu Deng...

Jul 23, 2026 — Deng is the 11th current or...



 University of Chicago

